

duces a 5-colouring of  $G - v$  in which  $u$  and  $w$  receive the same colour, that of  $v_P$  from our colouring of  $G/P$ . The neighbours of  $v$  in this colouring of  $G - v$  thus use at most four colours, and we can colour  $v$  with the fifth colour.  $\square$

As a backdrop to the two famous theorems above, let us cite another well-known result:

**Theorem 5.1.3.** (Grötzsch 1959)

*Every planar graph not containing a triangle is 3-colourable.*

## 5.2 Colouring vertices

How do we determine the chromatic number of a given graph? How can we *find* a vertex-colouring with as few colours as possible? How does the chromatic number relate to other graph invariants, such as average degree, connectivity or girth?

Straight from the definition of the chromatic number we may derive the following upper bound:

**Proposition 5.2.1.** *Every graph  $G$  with  $m$  edges satisfies*

$$\chi(G) \leq \frac{1}{2} + \sqrt{2m + \frac{1}{4}}.$$

*Proof.* Let  $c$  be a vertex colouring of  $G$  with  $k = \chi(G)$  colours. Then  $G$  has at least one edge between any two colour classes: if not, we could have used the same colour for both classes. Thus,  $m \geq \frac{1}{2}k(k-1)$ . Solving this inequality for  $k$ , we obtain the assertion claimed.  $\square$

*greedy  
algorithm*

One obvious way to colour a graph  $G$  with not too many colours is the following *greedy algorithm*: starting from a fixed vertex enumeration  $v_1, \dots, v_n$  of  $G$ , we consider the vertices in turn and colour each  $v_i$  with the first available colour – e.g., with the smallest positive integer not already used to colour any neighbour of  $v_i$  among  $v_1, \dots, v_{i-1}$ . In this way, we never use more than  $\Delta(G) + 1$  colours, even for unfavourable choices of the enumeration  $v_1, \dots, v_n$ . If  $G$  is complete or an odd cycle, then this is even best possible.

In general, though, this upper bound of  $\Delta + 1$  is rather generous, even for greedy colourings. Indeed, when we come to colour the vertex  $v_i$  in the above algorithm, we only need a supply of  $d_{G[v_1, \dots, v_i]}(v_i) + 1$  rather than  $d_G(v_i) + 1$  colours to proceed; recall that, at this stage, the algorithm ignores any neighbours  $v_j$  of  $v_i$  with  $j > i$ . Hence in most graphs,

there will be scope for an improvement of the  $\Delta + 1$  bound by choosing a particularly suitable vertex ordering to start with: one that picks vertices of large degree early (when most neighbours are ignored) and vertices of small degree last. Locally, the number  $d_{G[v_1, \dots, v_i]}(v_i) + 1$  of colours required will be smallest if  $v_i$  has minimum degree in  $G[v_1, \dots, v_i]$ . But this is easily achieved: we just choose  $v_n$  first, with  $d(v_n) = \delta(G)$ , then choose as  $v_{n-1}$  a vertex of minimum degree in  $G - v_n$ , and so on.

The least number  $k$  such that  $G$  has a vertex enumeration in which each vertex is preceded by fewer than  $k$  of its neighbours is called the *colouring number*  $\text{col}(G)$  of  $G$ . The enumeration we just discussed shows that  $\text{col}(G) \leq \max_{H \subseteq G} \delta(H) + 1$ . But for  $H \subseteq G$  clearly also  $\text{col}(G) \geq \text{col}(H)$  and  $\text{col}(H) \geq \delta(H) + 1$ , since the ‘back-degree’ of the last vertex in any enumeration of  $H$  is just its ordinary degree in  $H$ , which is at least  $\delta(H)$ . So we have proved the following:

colouring  
number  
 $\text{col}(G)$

**Proposition 5.2.2.** *Every graph  $G$  satisfies*

$$\chi(G) \leq \text{col}(G) = \max \{ \delta(H) \mid H \subseteq G \} + 1.$$

□

The colouring number of a graph is closely related to its arboricity; see Exercise 14 and the remark following Theorem 2.4.3.

Proposition 5.2.2 shows that every  $k$ -chromatic graph has a subgraph of minimum degree at least  $k - 1$ . In fact, it has a  $k$ -chromatic such subgraph:

**Lemma 5.2.3.** *Every  $k$ -chromatic graph has a  $k$ -chromatic subgraph of minimum degree at least  $k - 1$ .*

[7.3]  
[9.2.1]  
[9.2.3]  
[11.2.3]

*Proof.* Given  $G$  with  $\chi(G) = k$ , let  $H \subseteq G$  be minimal with  $\chi(H) = k$ . If  $H$  had a vertex  $v$  of degree  $d_H(v) \leq k - 2$ , we could extend a  $(k - 1)$ -colouring of  $H - v$  to one of  $H$ , contradicting the choice of  $H$ . □

As we have seen, every graph  $G$  satisfies  $\chi(G) \leq \Delta(G) + 1$ , with equality for complete graphs and odd cycles. In all other cases, this general bound can be improved a little:

**Theorem 5.2.4.** (Brooks 1941)

*Let  $G$  be a connected graph. If  $G$  is neither complete nor an odd cycle, then*

$$\chi(G) \leq \Delta(G).$$

*Proof.* We apply induction on  $|G|$ . If  $\Delta(G) \leq 2$ , then  $G$  is a path or a cycle, and the assertion is trivial. We therefore assume that  $\Delta := \Delta(G) \geq 3$ , and that the assertion holds for graphs of smaller order. Suppose that  $\chi(G) > \Delta$ .

Let  $v \in G$  be a vertex and  $H := G - v$ . Then  $\chi(H) \leq \Delta$ : by induction, every component  $H'$  of  $H$  satisfies  $\chi(H') \leq \Delta(H') \leq \Delta$  unless  $H'$  is complete or an odd cycle, in which case  $\chi(H') = \Delta(H') + 1 \leq \Delta$  as every vertex of  $H'$  has maximum degree in  $H'$  and one such vertex is also adjacent to  $v$  in  $G$ .

Since  $H$  can be  $\Delta$ -coloured but  $G$  cannot, we have the following:

*Every  $\Delta$ -colouring of  $H$  uses all the colours  $1, \dots, \Delta$  on the neighbours of  $v$ ; in particular,  $d(v) = \Delta$ .* (1)

Given any  $\Delta$ -colouring of  $H$ , let us denote the neighbour of  $v$  coloured  $i$  by  $v_i$ ,  $i = 1, \dots, \Delta$ . For all  $i \neq j$ , let  $H_{i,j}$  denote the subgraph of  $H$  spanned by all the vertices coloured  $i$  or  $j$ .

*For all  $i \neq j$ , the vertices  $v_i$  and  $v_j$  lie in a common component  $C_{i,j}$  of  $H_{i,j}$ .* (2)

Otherwise we could interchange the colours  $i$  and  $j$  in one of those components; then  $v_i$  and  $v_j$  would be coloured the same, contrary to (1).

*$C_{i,j}$  is always a  $v_i$ - $v_j$  path.* (3)

Indeed, let  $P$  be a  $v_i$ - $v_j$  path in  $C_{i,j}$ . As  $d_H(v_i) \leq \Delta - 1$ , the neighbours of  $v_i$  have pairwise different colours: otherwise we could recolour  $v_i$ , contrary to (1). Hence the neighbour of  $v_i$  on  $P$  is its only neighbour in  $C_{i,j}$ , and similarly for  $v_j$ . Thus if  $C_{i,j} \neq P$ , then  $P$  has an inner vertex with three identically coloured neighbours in  $H$ ; let  $u$  be the first such vertex on  $P$  (Fig. 5.2.1). Since at most  $\Delta - 2$  colours are used on the neighbours of  $u$ , we may recolour  $u$ . But this makes  $P \hat{u}$  into a component of  $H_{i,j}$ , contradicting (2).

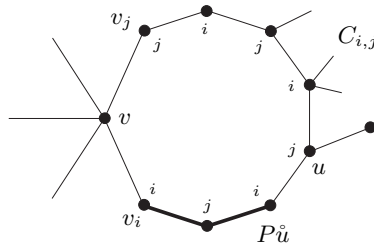


Fig. 5.2.1. The proof of (3) in Brooks's theorem

*For distinct  $i, j, k$ , the paths  $C_{i,j}$  and  $C_{i,k}$  meet only in  $v_i$ .* (4)

For if  $v_i \neq u \in C_{i,j} \cap C_{i,k}$ , then  $u$  has two neighbours coloured  $j$  and two coloured  $k$ , so we may recolour  $u$ . In the new colouring,  $v_i$  and  $v_j$  lie in different components of  $H_{i,j}$ , contrary to (2).

The proof of the theorem now follows easily. If the neighbours of  $v$  are pairwise adjacent, then each has  $\Delta$  neighbours in  $N(v) \cup \{v\}$  already, so  $G = G[N(v) \cup \{v\}] = K^{\Delta+1}$ . As  $G$  is complete, there is nothing to show. We may thus assume that  $v_1 v_2 \notin G$ , where  $v_1, \dots, v_\Delta$  derive their names from some fixed  $\Delta$ -colouring  $c$  of  $H$ . Let  $u \neq v_2$  be the neighbour of  $v_1$  on the path  $C_{1,2}$ ; then  $c(u) = 2$ . Interchanging the colours 1 and 3 in  $C_{1,3}$ , we obtain a new colouring  $c'$  of  $H$ ; let  $v'_i, H'_{i,j}, C'_{i,j}$  etc. be defined with respect to  $c'$  in the obvious way. As a neighbour of  $v_1 = v'_3$ , our vertex  $u$  now lies in  $C'_{2,3}$ , since  $c'(u) = c(u) = 2$ . By (4) for  $c$ , however, the path  $\hat{v}_1 C_{1,2}$  retained its original colouring, so  $u \in \hat{v}_1 C_{1,2} \subseteq C'_{1,2}$ . Hence  $u \in C'_{2,3} \cap C'_{1,2}$ , contradicting (4) for  $c'$ .  $\square$

$v_1, \dots, v_\Delta$   
 $c$   
 $u$   
 $c'$

We have so far seen some necessary conditions for high chromaticity, in the form of upper bounds on  $\chi$ . If  $\chi(G) \geq k$ , for example, then also  $\Delta \geq k$  (unless  $G$  is complete or an odd cycle), and  $G$  has a subgraph of minimum degree at least  $k-1$ . These conditions are far from sufficient, though: if  $G = K_{n,n}$ , say, they hold for all  $k \leq n$  but  $\chi(G) = 2$ .

It would be nice also to have some sufficient conditions for  $\chi \geq k$ . If they are easy to check, they might provide useful certificates for why we are unable to colour a given graph with few colours. If they could even be shown to be necessary too, they would ‘explain’ why certain graphs are highly chromatic – just as the marriage condition in Hall’s theorem ‘explains’ why certain matchings in bipartite graphs fail: its violation clearly prevents a graph from having the desired matching, and it is violated every time such a matching fails to exist.

For example, we might try to determine the class  $\mathcal{X}_k$  of  $\subseteq$ -minimal graphs that cannot be coloured with fewer than  $k$  colours. As is easy to check (cf. Lemma 12.6.1.), a given graph  $G$  satisfies  $\chi(G) \geq k$  if and only if it has a subgraph in  $\mathcal{X}_k$ , just as in Kuratowski’s planarity theorem with minors or topological minors. So containing any graph from  $\mathcal{X}_k$  is a certificate for  $\chi \geq k$ , and these certificates together ‘explain’ this phenomenon in the sense discussed.

(12.6.1)

But will these certificates be easy to find in an arbitrary  $k$ -chromatic graph, or at least easy to check? That is, will it be easy to verify that a given graph  $X \in \mathcal{X}_k$  is indeed in  $\mathcal{X}_k$ , or even just that  $\chi(X) \geq k$ ? We shall return to this question in a moment.

One obvious sufficient condition for  $\chi(G) \geq k$  is that  $K^k \subseteq G$ . But this condition is not necessary: as Theorem 5.2.5 will show,  $k$ -chromatic graphs need not even contain a triangle. Hence while  $K^k$  certainly lies in  $\mathcal{X}_k$ , it is not its only element. Conversely, Lemma 5.2.3 implies that all the graphs in  $\mathcal{X}_k$  have minimum degree at least  $k-1$ ; but not all graphs of minimum degree  $k-1$  are in  $\mathcal{X}_k$ , since they need not satisfy  $\chi \geq k$ .

The following theorem of Erdős implies that  $\mathcal{X}_k$  cannot be finite. In fact, it implies that for no  $k$  is there a finite set  $\mathcal{X}$  of graphs  $X$  with  $\chi(X) \geq 3$  such that every  $k$ -chromatic graph has a subgraph in  $\mathcal{X}$ :

[9.2.3]

**Theorem 5.2.5.** (Erdős 1959)

For every integer  $k$  there exists a graph  $G$  with girth  $g(G) > k$  and chromatic number  $\chi(G) > k$ .

Theorem 5.2.5 was first proved non-constructively using random graphs, and we shall give this proof in Chapter 11.2. Constructing graphs of large chromatic number and girth directly is not easy; cf. Exercise 22 for the simplest case.

The message of Erdős's theorem is that, contrary perhaps to what we had hoped, large chromatic number can occur as a purely global phenomenon: locally, around each vertex, a graph of large girth looks just like a tree, and in particular is 2-colourable there. But what exactly can cause high chromaticity as a global phenomenon remains a mystery.

Nevertheless, there exists a simple – though not always short – procedure to construct all the graphs of chromatic number at least  $k$ . For each  $k \in \mathbb{N}$ , let us define the class of  $k$ -constructible graphs recursively as follows:

$k$ -con-  
structible

- (i)  $K^k$  is  $k$ -constructible.
- (ii) If  $G$  is  $k$ -constructible and two vertices  $x, y$  of  $G$  are non-adjacent, then also  $(G + xy)/xy$  is  $k$ -constructible.
- (iii) If  $G_1, G_2$  are  $k$ -constructible and there are vertices  $x, y_1, y_2$  such that  $G_1 \cap G_2 = \{x\}$  and  $xy_1 \in E(G_1)$  and  $xy_2 \in E(G_2)$ , then also  $(G_1 \cup G_2) - xy_1 - xy_2 + y_1y_2$  is  $k$ -constructible (Fig. 5.2.2).

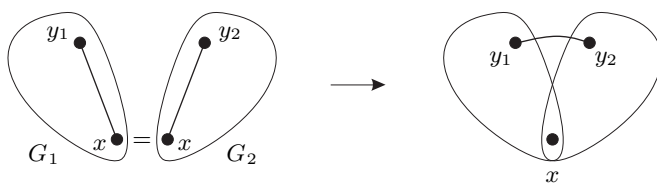


Fig. 5.2.2. The Hajós construction (iii)

One easily checks inductively that all  $k$ -constructible graphs – and hence their supergraphs – are at least  $k$ -chromatic. For example, any colouring of the graph  $(G + xy)/xy$  in (ii) induces a colouring of  $G$ , and hence by inductive assumption uses at least  $k$  colours. Similarly, in any colouring of the graph constructed in (iii) the vertices  $y_1$  and  $y_2$  do not both have the same colour as  $x$ , so this colouring induces a colouring of either  $G_1$  or  $G_2$  and hence uses at least  $k$  colours.

It is remarkable, though, that the converse holds too:

**Theorem 5.2.6.** (Hajós 1961)

Let  $G$  be a graph and  $k \in \mathbb{N}$ . Then  $\chi(G) \geq k$  if and only if  $G$  has a  $k$ -constructible subgraph.

*Proof.* Let  $G$  be a graph with  $\chi(G) \geq k$ ; we show that  $G$  has a  $k$ -constructible subgraph. Suppose not; then  $k \geq 3$ . Adding some edges if necessary, let us make  $G$  edge-maximal with the property that none of its subgraphs is  $k$ -constructible. Now  $G$  is not a complete  $r$ -partite graph for any  $r$ : for then  $\chi(G) \geq k$  would imply  $r \geq k$ , and  $G$  would contain the  $k$ -constructible graph  $K^k$ .

Since  $G$  is not a complete multipartite graph, non-adjacency is not an equivalence relation on  $V(G)$ . So there are vertices  $y_1, x, y_2$  such that  $y_1x, xy_2 \notin E(G)$  but  $y_1y_2 \in E(G)$ . Since  $G$  is edge-maximal without a  $k$ -constructible subgraph, each edge  $xy_i$  lies in some  $k$ -constructible subgraph  $H_i$  of  $G + xy_i$  ( $i = 1, 2$ ).

Let  $H'_2$  be an isomorphic copy of  $H_2$  that contains  $x$  and  $H_2 - H_1$  but is otherwise disjoint from  $G$ , together with an isomorphism  $v \mapsto v'$  from  $H_2$  to  $H'_2$  that fixes  $H_2 \cap H'_2$  pointwise. Then  $H_1 \cap H'_2 = \{x\}$ , so

$$H := (H_1 \cup H'_2) - xy_1 - xy'_2 + y_1y'_2$$

is  $k$ -constructible by (iii). One vertex at a time, let us identify in  $H$  each vertex  $v' \in H'_2 - G$  with its partner  $v$ ; since  $vv'$  is never an edge of  $H$ , each of these identifications amounts to a construction step of type (ii). Eventually, we obtain the graph

$$(H_1 \cup H_2) - xy_1 - xy_2 + y_1y_2 \subseteq G;$$

this is the desired  $k$ -constructible subgraph of  $G$ .  $\square$

Does Hajós's theorem solve our Kuratowski-type problem for highly chromatic graphs, which was to find a class of graphs of chromatic number at least  $k$  with the property that every such graph has a subgraph in this class? Formally it does, albeit with an infinite characterizing class: the class of  $k$ -constructible graphs, which contains  $\mathcal{X}_k$ . Unlike Kuratowski's characterization of planar graphs, however, this does not – at least not obviously – make Hajós's theorem a good characterization of the graphs of chromatic number  $< k$ : as one can show, proving that a given  $k$ -constructible graph is indeed  $k$ -constructible is just as hard as proving that a graph of chromatic number  $\geq k$  does indeed need at least  $k$  colours. See the notes for details.

## 5.3 Colouring edges

Clearly, every graph  $G$  satisfies  $\chi'(G) \geq \Delta(G)$ . For bipartite graphs, we have equality here: